Dylan T. Casey

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Burlington, VT 05405
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EDUCATION

PhD in Complex Systems and Data Science — July 2018 – present
University of Vermont, Burlington, VT
GPA: 3.70

BS in Mathematical Sciences — September 2012 – May 2017
University of Vermont, Burlington, VT
GPA: 3.36 — Major GPA: 3.59
GRE — Q: 167(92%) V: 151(52%) W: 4.5(82%)

EXPERIENCE

Research Lab Technician — May 2017 – July 2018
Lab Assistant — September 2014 – May 2017
Janssen-Heininger Lab, Department of Pathology and Laboratory Medicine
University of Vermont, Burlington, VT
- Work-study during school year

Lab Technician — June 2016 – August 2016
David Chapman, Ph.D., Irvin Lab, Department of Medicine, Pulmonary
University of Vermont, Burlington, VT
- Time split with Janssen-Heininger Lab

AWARDS & FUNDING

Vermont Lung Center (VLC) predoctoral fellow (T32 HL-076122)
NSF Graduate Research Fellowship 2020 — Honorable Mention
Dean's List — Fall 2014, Fall 2016

LAB SKILLS

Managing lab, Lab Safety Officer, and administrative duties
Managing mouse colony (breeding multi-transgenic strains, genotyping, etc.)
Tissue staining, Microscopy, and histological analysis (MetaMorph)
Performing/analyzing various assays (ELISA, Bio-Rad, qPCR, hydroxyproline, etc.)
Flexivent (lung mechanics)
Micro-CT

COMPUTER SKILLS

Languages: Expert in MATLAB; Proficient in Python, R, LaTeX

TEACHING & MENTORING

Teaching assistant
MATH 300: Principles of Complex Systems, Vol. 1 — FALL 2022
MATH 303: Principles of Complex Systems, Vol. 2 — SPRING 2023

Undergraduate Thesis
Will Wright (*BS in Biomedical engineering*) — May 2023

TALKS

VLC seminar
03/08/22: *Local fractal dimension variation of collagen structure in IPF*
01/11/22: *Computational models of pulmonary fibrosis*
8/10/21: *Modeling the mechanical behavior of the fibrotic alveolar wall*

Am I A Seminar? (graduate student math seminar)

3/30/17: *The Hardy-Littlewood Maximal Function*

9/26/16: *The Dirichlet Problem: An Introduction to Fourier Series*

MANUSCRIPTS

Bou Jawde S, Karrobi K, Roblyer D, Vicario F, Herrmann J, **Casey DT**, et al. **Inflation instability in the lung: an analytical model of a thick-walled alveolus with wavy fibres under large deformations** *J R Soc Interface.* 2021 Oct 18;(183):20210594. doi: 10.1098/rsif.2021.0594.

Casey DT, Bou Jawde S, Herrmann J, et al. **Percolation of collagen stress in a random network model of the alveolar wall**. *Sci Rep.* 2021 Aug 17;11:16654. doi: 10.1038/s41598-021-95911-w.

Chapman DG, **Casey DT**, Ather JL, et al. **The Effect of Flavored E-cigarettes on Murine Allergic Airways Disease**. *Sci Rep.* 2019 Sep 20;9:13671. doi: 10.1038/s41598-019-50223-y.

Anathy V, Lahue KG, Chapman DG, Chia SB, **Casey DT**, et al. **Reducing protein oxidation reverses lung fibrosis**. *Nat Med.* 2018 Aug;24(8):1128-1135. doi: 10.1038/s41591-018-0090-y.

Qian X, Aboushousha R, van de Wetering C, Chia SB, Amiel E, Schneider RW, van der Velden JL, Lahue KG, Hoagland DA, **Casey DT**, et al. **Interleukin-1β/inhibitory kappa B kinase epsilon-induced glycolysis augment epithelial effector function and promote allergic airways disease**. *J Allergy and Clin Immunol.* 2017 Nov 3. pii: S0091-6749(17)31663-9. doi:10.1016/j.jaci.2017.08.043

McMillan DH, van der Velden JL, Lahue KG, Qian X, Schneider RW, Iberg MS, Nolin JD, Abdalla S, **Casey DT**, et al. **Attenuation of lung fibrosis in mice with a clinically relevant inhibitor of glutathione-S-transferase π**. *JCI Insight.* 2016 Jun 2;1(8). pii. e85717. doi:10.1172/jci.insight.85717.

RECENT POSTERS

Casey DT, Mori V, Herrmann J, et al. **Three-dimensional spatial organization of fibrotic tissue: an agent-based model**. Poster presented at: American Thoracic Society Conference 2022; May 17, 2022; San Francisco, CA.

Casey DT, Mori V, Herrmann J, et al. **Rich-get-richer mechanism in fibrosis mimics fractal collagen structure**. Poster presented remotely at: Biomedical Engineering Society Conference 2021; October 12-15, 2021; Orlando, FL.

REFERENCES

Jason H.T. Bates, Ph.D., D.Sc.
Professor of Medicine (primary)
Professor of Electrical and Biomedical Engineering (secondary)
Jason.H.Bates@med.uvm.edu

Matthew Poynter, Ph.D.
Professor of Medicine
Associate Director of VLC
matthew.poynter@uvm.edu

Peter S. Dodds, Ph.D.
Professor of Mathematics and Statistics
Director of Vermont Complex Systems Center
peter.dodds@uvm.edu